

Abstract No.

Local calvarial myeloid reservoirs do not contribute to tumor associated macrophages in Glioblastoma

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Recent evidence shows that monocyte-derived macrophages (MDMs) invading into the brain can have a dual source: the adjacent skull bone marrow via direct dural vascular channels and the systemic blood circulation. The MDMs can seed the brain parenchyma under acute neuroinflammation and infiltrate the microglia-free brain.

Although monocyte-derived macrophages (MDMs) represent a major subset of tumor-associated macrophages in glioblastoma (GBM), local calvarial marrow is increasingly recognized as an active source of immune cells during CNS malignancy, supplying cells of the adaptive immune system or neutrophils to the tumor. Whether skull-derived macrophages are also actively recruited to the GBM microenvironment, however, remains unclear. To track the specific contribution of skull bone marrow-derived macrophages during tumor growth, we performed calvarial skull flap transplantation from Cx3cr1-GFP donor mice into Cx3cr1::CreERT2;PU.1fl/fl recipient mice. Conditional PU.1 knockout in microglia and macrophages cells induced a strong (95%) and long-lasting reduction in CNS myeloid cell populations. Into the same model we orthotopically implanted GBM cells. We analysed the mouse brains using confocal microscopy to visualize and quantify donor skull-derived Cx3cr1-GFP cell infiltration.

Surprisingly, we didn't observe skull bone marrow-derived MDMs in GBM. This lack of engraftment was consistent across all examined areas. These findings indicate that the MDMs recruited to glioblastoma arise dominantly from the systemic blood circulation, rather than from the adjacent skull marrow. This finding suggests a distinct tumor-driven recruitment program that bypasses the local calvarial bone marrow.



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